

14.2 Interpreting Joint and Marginal Frequencies

Lesson Introduction

 Benchmarks	MA.912.DP.3.1 Construct a two-way frequency table summarizing bivariate categorical data. Interpret joint and marginal frequencies, and determine possible associations in terms of a real-world context. MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data, and on whether it is univariate or bivariate. MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate, and interpret the different components and quantities in the display.			 Learning Targets	- I can determine joint and marginal frequencies. - I can interpret joint and marginal frequencies in a real-world context. - I can interpret a graphical display for bivariate categorical data.
 Benchmark Coherence	Building On		Working Toward		
	N/A		N/A		
 Essential Questions	- How can you determine joint and marginal frequencies? - How can you interpret joint and marginal frequencies in a real-world context? - What can you interpret from a graphical display for bivariate categorical data?				
 Lesson Narrative	In this lesson, students continue to explore two-way frequency tables. Student understanding of joint and marginal frequencies should be solidifying at this point in the unit, and they should be comfortable interpreting two-way frequency tables by the conclusion of Lesson 2.				
 Component Summary	14.2.1 Warm-Up (~5 min.)	Students complete two-way frequency table using information from a written narrative (<i>SE p. 311</i>)	14.2.3 Try It (10 min.)	Students complete a two-way frequency table and use it to answer questions (<i>SE p. 313</i>)	
	14.2.2 Collaboration (15-20 min.)	Complete two-way frequency table, answer, and write questions based on the information (<i>SE p. 312</i>)	14.2.4 Your Turn! (10 min.)	Students complete two-way frequency tables, determine, and interpret joint and marginal frequencies (<i>SE p. 314</i>)	
	14.2.5 Wrap-Up (~5 min.)	Ticket out the Door where the students find and interpret joint and marginal frequencies (<i>SE p. 315</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Joint Frequency - Marginal Frequency		(Optional) Two different colored highlighters		(Optional) Consider creating a chart for students to create a list of joint and marginal frequency questions for 14.2.2 Collaboration.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle with finding missing values in the two-way frequency tables. - Students may confuse joint and marginal frequencies.	- Gathering information from a real-world context and putting the information into a two-way frequency table can be a struggle for some students. Consider using a reading strategy to help students pull out and place the needed information. - During 14.2.3 Try It, consider having students highlight what is being asked for the questions in problem 2 to help students focus on what they are looking for.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Co-Craft Questions and Problems During 14.2.2 Collaboration, students work with a partner to write questions that can be answered with the data in the two-way frequency table with a focus on one question that can be answered with a joint frequency, and one that can be answered with a marginal frequency.	MA.K12.MTR.7.1: Real-World Contexts Clarifications explain that “mathematicians who apply mathematics to real-world contexts use models and methods to understand, represent, and solve problems.” Throughout this lesson students will create two-way frequency tables to model data from a real-world context. Then, use these models to help them interpret the data.
 Differentiate Instruction	For 14.2.2 Collaboration, consider creating a chart where students can write questions they created with their partner for joint and marginal frequencies. Then, as a class, check to see if the questions written work for each category. Displaying these questions allows for visual learners to see why or why not a question works for either joint or marginal frequencies	
	Joint Frequency Questions	Marginal Frequency Questions
Lesson Notes:		

14.2.1 Warm-Up: Bell Work

Component Overview: 14.2.1 Warm-Up provides the opportunity for students to revisit what was learned in the prior lesson. Allow students to read the narrative, then complete the two-way frequency table.



14.2.1 Warm-Up: Bell Work

1. A survey conducted at Ambidextrous High School asked all 1,700 students to indicate their grade level and if they are left-handed or right-handed. Only 59 of the 491 freshmen are left-handed. Out of the 382 students in the sophomore class, 289 of them are right-handed. There are 433 students in the junior class and 120 of them are left-handed. There are 307 right-handed seniors. Complete the two-way frequency table to display the results of the survey.



This Bell Work is meant to check student understanding of constructing a two-way frequency table from the previous lesson. During this lesson students continue to construct two-way frequency tables, but also interpret the joint and marginal frequencies.

Dominant Hand Survey

	<i>Freshmen</i>	<i>Sophomore</i>	<i>Junior</i>	<i>Senior</i>	Total
Right-Handed	432	289	313	307	1,341
Left-Handed	59	93	120	87	359
Total	491	382	433	394	1,700

14.2.2 Collaboration: Joint and Marginal Frequencies

Component Overview: This student-led Collaboration provides students with multiple opportunities to make sense of information as they complete a two-way frequency table and answer questions about joint, marginal, and joint relative frequencies using the table.



14.2.2 Collaboration: Joint and Marginal Frequencies

1. In 2016, the Summer Olympics were held in Rio de Janeiro, Brazil. The United States of America won the most medals with a total of 121. China came in second, winning 70 total medals. This was a much higher total than the 9 medals China won two years later in the 2018 Winter Olympics held in Pyeongchang, South Korea. Norway led the medal count in 2018 with 39, a much higher number than the 4 total medals Norway won two years earlier in Brazil. Canada came in third in the 2018 Winter Olympics, winning 29 total medals.
 - a. Work with your partner to use the data regarding medal counts from the 2016 Summer Olympics and the 2018 Winter Olympics to make a two-way frequency table showing how the different countries' medal counts compare to each other.

	U.S.	China	Norway	Canada	Total
2016 Summer Olympics	121	70	4	22	217
2018 Winter Olympics	23	9	39	29	100
Total	144	79	43	51	317

- b. How many total medals did the U.S. win in the 2016 and 2018 Olympics combined?

The U.S. won 144 total medals in the 2016 and 2018 Olympics combined.



Students should be collaborating throughout the component. Monitor and listen for common student misconceptions during this activity.



For students who are struggling, provide two different colored highlighters, one color for 2016, and another for 2018. Have students first read the narrative to highlight each country's medal count for 2016. Then, using the other color, have students read the narrative again to identify the medals won by each country for 2018.



Ensure that students have completed the two-way frequency table correctly before they move onto questions 1b – f.

14.2.2 Collaboration: Joint and Marginal Frequencies (continued)

- c. Which team won the most medals in the two Olympics combined?
The U.S. won the most medals in the two Olympics combined.
- d. How many medals did Norway win in the 2018 Olympics?
Norway won 39 medals in the 2018 Olympics.
- e. Complete the statement.

The U.S. won the most medals in either the 2016 or 2018 Olympics, winning 121 medals in those games.

- f. What does the joint relative frequency, $\frac{121}{317}$, mean in this context?
In this context, the joint relative frequency, $\frac{121}{317}$, means that the U.S. won 121 medals in the 2016 Summer Olympics out of the total medals won in the 2016 and 2018 Olympics combined.

2. Work with your partner to write questions that can be answered with the data in the table.
- a. Write at least one question that must be answered with a joint frequency.
Answers will vary. Which country won the least medals in the 2016 Olympics? How many medals did they win? Explain why this may have happened.
- b. Write at least one question that must be answered with a marginal frequency.
Answers will vary. Did these four countries win more medals in the 2016 or 2018 Olympics? How many more medals did these countries win in the 2016 Olympics than the 2018 Olympics?



Students may be confused by question 1e, because of the use of the word or when referring to the 2016 or 2018 Olympics. Students could also put that Norway won the most medals, winning 39 in those games if they are looking at the 2018 Olympics.



Consider creating a chart where students can write questions they created with their partner for joint and marginal frequencies. Then, as a class, check to see if the questions written work for each category. Displaying these questions allows for visual learners to see why or why not a question works for either joint or marginal frequencies.

Joint Frequency Questions	Marginal Frequency Questions

14.2.3 Try It: Mono Lake

Component Overview: Students get the opportunity to create a two-way frequency table to help them understand how the changing salinity of a lake has affected the organisms that live there. This is a student-led component where students are allowed to work with their partner.



14.2.3 Try It: Mono Lake

In California, there is a saltwater lake called Mono Lake. The salinity, or the concentration of salt in the lake, has varied over time. According to the Global Nature Fund, the salinity of Mono Lake was at an all-time high of 99 grams per liter (g/l) in 1982. The concentration of salt in Mono Lake increased as a result of water diversion. Streams that feed Mono Lake were diverted, and the fresh water was used for drinking water.

Regulations and compromises have helped control water diversion at Mono Lake since then, and the salinity of the lake is slowly returning to normal. In 2008, there were 79.8 grams of salt per liter in Mono Lake. As a result, the populations of two main organisms that live in the lake, brine shrimp and alkali flies, have increased to more optimal levels. In 1982, there were an estimated 2,000 brine shrimp per liter of lake water. In 2008, there were 3,000 brine shrimp per liter of lake water. In 1982, there were 1,200 alkali flies per liter of lake water. That number nearly tripled to 3,500 alkali flies per liter in 2008 when the salinity of Mono Lake was 79.8 g/l.

1. Construct a two-way frequency table that shows the alkali flies per liter of lake water and brine shrimp per liter of lake water living at Mono Lake when the salinity was 99 g/l and 79.8 g/l.

	Salinity of 99 g/l	Salinity of 79.8 g/l	Total
Brine Shrimp	2,000	3,000	5,000
Alkali Flies	1,200	3,500	4,700
Total	3,200	6,500	9,700



For students who struggle with filling out the table because the context switches from year to salinity, it may be helpful for students to make the connection here to 99 g/l to 1982, and 79.8 g/l to 2008 by writing the year about each on the two-way frequency table.

14.2.3 Try It: Mono Lake (continued)

2. Work with your partner to answer each question.

- a. How many alkali flies per liter of lake water were found in Mono Lake when the salinity was 99 g/l?

When the salinity of Mono Lake was 99 g/l there were 1,200 alkali flies per liter of lake water from Mono Lake.

- b. When the salinity of Mono Lake was 79.8 g/l, how many brine shrimp and alkali flies per liter of lake water were found?

When the salinity of Mono Lake was 79.8 g/l there were 6,500 brine shrimp and alkali flies per liter of lake water from Mono Lake.

- c. Which is greater — the combined total of alkali flies, or the combined total of brine shrimp in Mono Lake from these two samples?

From these two samples, the combined total of brine shrimp in Mono Lake is greater than the combined total of alkali flies.

- d. What was the salinity of Mono Lake when the population of alkali flies was 3,500 per liter?

When the population of alkali flies was 3,500 per liter, the salinity of Mono Lake was 79.8 g/l.

- e. What was the total amount of brine shrimp found in Mono Lake from the two studies?

The total amount of brine shrimp found in one liter of water from Mono Lake in 1982 and another liter from Mono Lake in 2008 is 5,000.

- f. Was the population of brine shrimp smaller when the salinity of Mono Lake was 99 g/l or 79.8 g/l?

The population of brine shrimp in Mono Lake was smaller when the salinity of Mono Lake was 99 g/l.



For students that finish early, consider having them go back and identify joint and marginal frequency questions.



Facilitate a conversation by prompting student responses to questions 2a – f. Ask students to clarify how they found their answers to the questions, so that students can hear how other students found their answers as they check their own answers.

14.2.4 Your Turn!

Component Overview: During 14.2.4 Your Turn!, students will practice completing a two-way frequency table and interpreting the joint and marginal frequencies from the context.



14.2.4 Your Turn!

- At Dupont Middle School in Jacksonville, FL, 635 students were asked which social media site they updated the most on a daily basis. Seventy-five seventh graders said they used Instagram the most. A total of 321 students said they updated Facebook the most. Of the 247 eighth graders, 23 of them updated Instagram the most. There are 181 sixth graders at Dupont. Of the 158 Twitter users, 45 of them were sixth graders. There were 123 eighth graders using Facebook.

Complete the two-way table summarizing the data.

Dupont Middle School Social Media Preference

	Facebook	Instagram	Twitter	Total
6 th	78	58	45	181
7 th	120	75	12	207
8 th	123	23	101	247
Total	321	156	158	635



To encourage mathematical reasoning as students complete the two-way frequency table, ask questions like:

- How do you use the information that is provided to find the missing values in the table?
- Where do you think a good starting point would be when filling in missing information?

14.2.4 Your Turn! (continued)

2. Austin studies birth weight and height as part of his research at Florida School of Traditional Midwifery in Gainesville, Florida. The following two-way table shows data from births in the month of June 2021 at UF Health Shands Hospital.

Births from UF Health Shands Hospital (June 2021)

	Less Than 5 lbs.	5-8 lbs.	More Than 8 lbs.	Total
Less Than 16 in.	21	9	2	32
16-20 in.	14	<u>97</u>	68	179
More Than 20 in.	0	8	31	39
Total	35	114	101	250

Use the table to answer the following questions and prompts.

- Circle the smallest marginal frequency in the table.
The smallest marginal frequency in the table is 32, the total of all babies born at UF Health Shands Hospital in June 2021 who were less than 16 inches in length.
- Underline the largest joint frequency in the table.
The largest joint frequency in the table is 97, the number of babies born at UF Health Shands Hospital in June 2021 who were 16 – 20 inches long and 5 – 8 pounds.
- Describe what the value 14 means in context of the data.
The number 14 in the table tells us that 14 babies born at UF Health Shands Hospital in June 2021 were 16 – 20 inches long and weighed less than 5 pounds.
- Describe what the value 101 means in context of the data.
The value 101 in the table tells us that 101 babies born at UF Health Shands Hospital in June 2021 weighed more than 8 pounds.



Prompt students to complete the two-way frequency table before answering questions 2a – d.



Students may only state that 14 babies born at UF Health Shands Hospital in June 2021 were less than 5 pounds, or that they were between 16 – 20 inches long. Ensure that both are included in their description.



If students finish early, have students identify if the questions for 2c and 2d are for a joint or marginal frequency.

14.2.5 Wrap-Up: Ticket Out the Door

Component Overview: Students complete a Ticket Out the Door where they identify joint and marginal frequencies, and interpret their meaning from the context.



14.2.5 Wrap-Up: Ticket Out the Door

Use the two-way table showing students' field trip preferences to answer the questions.

Students' Field Trip Preferences

	Amusement Park	Water Park	Mystery Destination	Total
Freshmen	25	9	20	54
Sophomores	35	6	5	46
Total	60	15	25	100

- Underline the smallest joint frequency.
- Explain what the smallest joint frequency represents in terms of students' field trip preferences.
The smallest joint frequency of 5 represents the number of sophomores who want to go to a mystery destination for the field trip.
- Circle the largest marginal frequency.
- Explain what the largest marginal frequency represents in terms of students' field trip preferences.
The largest marginal frequency of 60 represents the number of freshman and sophomores who want to go to the amusement park for the field trip.



Consider using the data provided from 14.2.4 Your Turn! and 14.2.5 Wrap-Up to form small groups for Lesson 3. The information provided from both can help determine students that struggle with determining joint and marginal frequencies, and/or interpreting joint and marginal frequencies.



Consider our online resources for additional remediation opportunities for students to practice or test their abilities.